

Hawkeye EQMS · Deviation / Non-Conformance · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

Contents

- 1. Module overview
- 2. Standard pharma EQMS vs Hawkeye — gap analysis
- 3. Personas + role mapping
- 4. Feature catalogue (click-by-click)
- 5. Full lifecycle walkthrough — one transaction, end-to-end
- 6. AI assistance map
- 7. Test results matrix
- 8. Known gaps + roadmap

1. Module overview

Module	Deviation / Non-Conformance
Pharma purpose	Capture, investigate, disposition and close any planned or unplanned deviation from a specification, SOP, batch record, or quality-system requirement. Drive a CAPA when the deviation is systemic.
Compliance basis	ISO 9001:2015 §10.2 (non-conformity & corrective action) · 21 CFR 211.192 (production record review · investigations) · ICH Q10 §3.2.3 (deviation management) · EU GMP Chapter 1.4 (quality system).
Hawkeye routes	/nonconformance (frontend) · /api/deviations (backend)
Hawkeye model file	backend/src/models/DeviationModel.js
Tenant module flag	modules.EVENT_MANAGEMENT

A 'deviation' is any departure from the approved batch record, validated procedure, or specification. Pharma QA teams treat the deviation as the entry-point of the quality investigation lifecycle: it determines whether the affected batch can be released, whether a CAPA is required, and (if patient-safety / regulatory exposure) whether a Field Alert Report or recall must be filed. Hawkeye stores deviations in the `deviations` collection, exposes them to users with the `EVENT_MANAGEMENT` module flag enabled, and offers AI assistance at the investigation step (Wave 1 · 5-Why scaffolder + CAPA RCA drafter) and at the disposition step (Wave 3 · predictive CAPA outcome).

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Single intake form for any operator to log a deviation 21 CFR 211.192	+ Report Deviation dialog on /nonconformance with title, description, type (PLANNED / UNPLANNED), classification (CRITICAL / MAJOR / MINOR), category (10 enums), area, processStep, productName, immediateActions	MET	

Pharma expectation	Hawkeye implementation	Status	Note
Auto-numbering of deviation records (DEV-YYYY-NNNN) GMP traceability	deviationNumber auto-generated server-side (DeviationModel.js:54). Format DEV-YYYY-NNNN.	MET	
Multi-step lifecycle with role gates (operator → QA → Head of QA → VP) ICH Q10 §3.2.3	9-state machine (REPORTED → UNDER_ASSESSMENT → UNDER_INVESTIGATION → PENDING_DISPOSITION → PENDING_CAPA_DECISION → CAPA_REQUIRED → PENDING_CLOSURE → CLOSED, plus CANCELLED). Transitions gated by EDITOR_ROLES.	MET	
Impact assessment captured before investigation (product / patient / batch / regulatory) 21 CFR 211.192	POST /assess endpoint requires productQualityImpact, patientSafetyImpact, batchDisposition, regulatoryImpact (deviationRoutes.js:104-124)	MET	
Root-cause analysis with method tracking (5-Why / Fishbone / FTA / FMEA / Pareto) ICH Q9	POST /investigate accepts method enum + summary + rootCause + rootCauseCategory. AI 5-Why scaffolder available in the View+AI drawer.	MET	
Batch disposition decision (RELEASE / REJECT / REWORK / REPROCESS / QUARANTINE) 21 CFR 211.165	POST /dispose with decision + justification. Five-value enum.	MET	
Decision whether CAPA is required, with auto-creation of CAPA record ISO 9001 §10.2.1	POST /capa-decision with capaRequired flag. autoCreateCapa=true triggers crossModuleService.createCapaFromDeviation() — produces a real capa-v2 record, not just a flag.	MET	
Closure with electronic signature meeting 21 CFR Part 11 21 CFR Part 11 §11.50	POST /close captures closureNotes + closedBy + closedAt. Generic /api/electronic-signatures/sign endpoint exists and can be wired to closure but is NOT enforced today.	PARTIAL	Wire e-sig requirement to closure (1-line check)
Linked records: CAPAs, complaints, change controls, audits ICH Q10 §3.2.5	linkedCAPAIds, linkedComplaintIds, linkedChangeControlIds, linkedAuditIds, linkedDeviationIds arrays on the model. Cross-module endpoint POST /api/quality/capa-from-deviation creates + links.	MET	
Field Alert Report / MDR reportability flag 21 CFR 314.81 / 21 CFR 803	regulatoryImpact captured as free-text on the assessment but no MDR / FAR boolean / deadline timer.	GAP	Add isFieldAlertReportable + farDueDate + auto-reminder
Trend / signal detection across deviations (e.g. 3 OOS in 30 days on same equipment) ICH Q10 §3.2.4 (KPI monitoring)	Wave 3 SignalAlertsList — Z-score cluster detector. Live alert today: equipment:NVX-PRESS-001 · size 3 · z=3.4.	MET	
AI assistance to accelerate RCA + draft CAPA + predict outcome FDA AI guidance Jan 2025	View + AI drawer offers Scaffold 5-Why (Wave 1) + Draft CAPA RCA (Wave 1) + Predictive CAPA badge (Wave 3) + Create CAPA from RCA cross-module button.	MET	
Audit trail of every state change + every AI accept/reject (ALCOA+) 21 CFR Part 11 §11.10(e)	auditTrailModel logs deviation transitions. AI outputs logged to AiActionMetric + DataIntegrityLog with prompt-version + retrieval-set hashes.	MET	
Per-tenant module enable / disable Multi-tenant SaaS	tenantModuleConfig with EVENT_MANAGEMENT boolean. Frontend hides nav + 403s when disabled.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Kenji Tanaka QA Specialist (admin)	qa.specialist@novex-pharma.demo	Reports deviations from the floor, runs the initial impact assessment, drives investigation + RCA, accepts AI-drafted RCA, creates CAPA from drawer.	REPORTED UNDER_ASSESSMENT UNDER_INVESTIGATION
James Thompson Head of QA (admin)	qa.head@novex-pharma.demo	Reviews investigation conclusions, makes batch disposition decision, decides whether systemic CAPA is required, signs closure.	PENDING_DISPOSITION PENDING_CAPA_DECISION CAPA_REQUIRED PENDING_CLOSURE CLOSED
Elena Vasquez VP Quality (tenant_admin)	vp.quality@novex-pharma.demo	Final approver on CRITICAL deviations; reviews closure on the management-review cycle. Signs e-sig where enforced.	PENDING_CLOSURE CLOSED
Marcus Brown Regulatory Affairs (admin)	regulatory@novex-pharma.demo	Determines MDR / Field Alert reportability when deviation has patient-safety or regulatory exposure.	UNDER_ASSESSMENT UNDER_INVESTIGATION
Maria Santos Lead Auditor (auditor)	audit.lead@novex-pharma.demo	Reviews deviation closure during periodic internal audit. Triage CAPAs spawned from deviations.	CLOSED (audit-time review)

4. Feature catalogue (click-by-click)

4.1 View deviation register

What it does	Land on /nonconformance to see every deviation in the tenant with classification + status chips and inline workflow buttons.
Menu / route	Top nav · 'Deviations' tile · /nonconformance
Available to roles	admin buyer auditor tenant_admin
Backend API	GET /api/deviations?page=1&limit=20

CLICK-BY-CLICK

- NAVIGATE** Click 'Deviations' in the top nav (or open /nonconformance directly)
 - Page loads with header 'Deviation / NC Manager' + the counter chips '3 total · 3 open' (using seeded data)
- WAIT** Wait for the loading spinner to disappear (data fetched from /api/deviations)
 - Three rows render: DEV-DEMO-001 OOS dissolution, DEV-DEMO-002 calibration drift, DEV-DEMO-003 viable contamination

walkthrough/02-kenji.png

Tip: If the list is empty after seeding, you're hitting the wrong backend. Confirm APP_API_BASE_URL on the Vercel frontend points to https://hawkeye-backend-dev.vercel.app (the v2.0 fix).

4.2 Filter the register

What it does	Narrow the list by classification, status, or category.
Menu / route	/nonconformance – filter row above the deviation list
Available to roles	any viewer

CLICK-BY-CLICK

- 1. **CLICK** Click the 'Classification' dropdown
 - Dropdown shows CRITICAL / MAJOR / MINOR
- 2. **CLICK** Click 'Status' dropdown
 - Dropdown shows the 9 lifecycle states
- 3. **CLICK** Click 'Category' dropdown
 - 10 categories: Process, Equipment, Material, Documentation, Environmental, Laboratory, Packaging, Storage, Personnel, Other

4.3 Report a new deviation (intake dialog)

What it does	Open the +Report Deviation modal, fill in the intake form, save. Creates a record in REPORTED state.
Menu / route	/nonconformance – top-right red '+' Report Deviation' button
Available to roles	admin buyer auditor tenant_admin inspector workflow_manager
Backend API	POST /api/deviations

CLICK-BY-CLICK

- 1. **CLICK** Click '+ Report Deviation' (top-right)
 - Dialog opens with 8 form fields
- 2. **TYPE** Fill 'Title' — short headline (e.g. 'OOS dissolution on batch NVX-2026-B015')
 - Required; submit disabled until non-empty
- 3. **TYPE** Fill 'Description' — multi-line narrative
 - Required; submit disabled until non-empty
- 4. **CLICK** Choose 'Type' — UNPLANNED (default) or PLANNED
 - Most production deviations are UNPLANNED
- 5. **CLICK** Choose 'Category' — Process, Equipment, Material, etc.
 - Drives downstream KPI grouping
- 6. **CLICK** Choose 'Classification' — CRITICAL / MAJOR / MINOR (default MINOR)
 - CRITICAL gates VP-level closure
- 7. **TYPE** Fill 'Department' / 'Area' / 'Process Step' / 'Product Name' / 'Immediate Actions'
 - All optional — but stronger investigations capture them
- 8. **CLICK** Click 'Save' (red button)
 - Dialog closes; new row appears at top of list with status REPORTED

FIELDS CAPTURED

Field	Required	Validation / values	Notes
title	Yes	string	max ~200 chars practical
description	Yes	multi-line text	captured in /investigate later
deviationType	No	PLANNED UNPLANNED	default UNPLANNED
category	No	PROCESS EQUIPMENT MATERIAL DOCUMENTATION ENVIRONMENTAL LABORATORY PACKAGING STORAGE PERSONNEL OTHER	

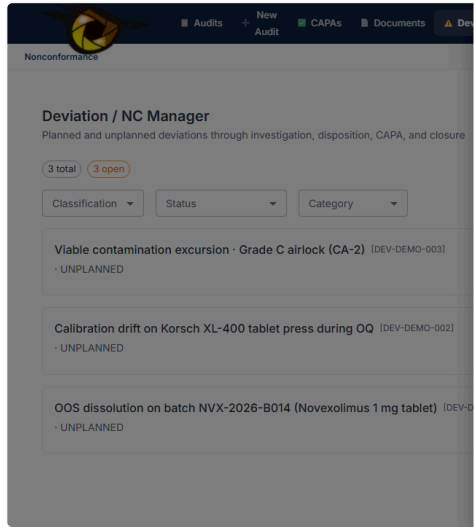
Field	Required	Validation / values	Notes
classification	No	CRITICAL MAJOR MINOR	default MINOR
productName	No	string	
batchNumbers	No	array of strings	
immediateActions	No	string	what was done at the moment of detection (quarantine, line stop, ...)

4.4 View + AI drawer (per row)

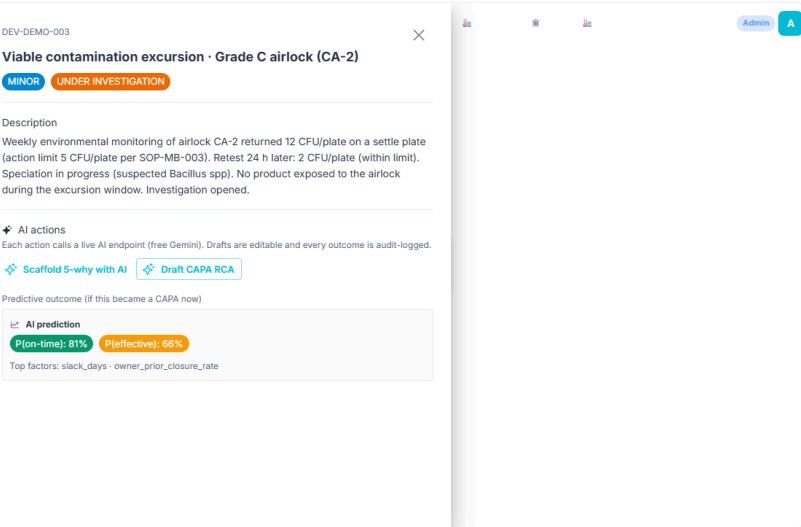
What it does	Right-side drawer opens when you click 'View + AI' on any deviation row. Shows full narrative + the three AI assists.
Menu / route	/nonconformance – primary blue 'View + AI' button on every row (data-testid='deviation-view-ai')
Available to roles	any role with row visibility
AI assistance	AI Predictive CAPA badge auto-renders on every drawer open.

CLICK-BY-CLICK

- CLICK** Click the blue 'View + AI' button on any deviation row
→ Drawer slides in from the right (640 px wide). Shows DEV-XXX header, classification + status chips, full description, AI actions section, AI prediction card.
- WAIT** Predictive CAPA card auto-loads (POST /api/ai/predict/capa-outcome)
→ Two green/orange chips: P(on-time)=NN% · P(effective)=NN% · 'Top factors: ...'. ~700 ms



walkthrough/03-kenji.png



4.5 AI · Scaffold 5-Why

What it does	Inline AI agent that turns the deviation narrative into a 5-level Why chain with citations and 6M (Man/Machine/Method/Material/Measurement/Environment) categorisation.
Menu / route	View + AI drawer · button labelled 'Scaffold 5-why with AI'
Available to roles	any with drawer access
Backend API	POST /api/ai/deviation/scaffold-five-why
AI assistance	AI Free Gemini 2.5 Flash-Lite. Citation-gated; falls back if confidence < 0.4.

CLICK-BY-CLICK

- CLICK** Click 'Scaffold 5-why with AI'

→ Popover opens with a spinner; AI request fires

2. **WAIT** Wait ~2-4 s for the AI response

→ Popover renders 5 numbered Why steps with answer + citation refs (e.g. SOP-QC-014:3.2). Below: 6M categorisation, 6 follow-up questions.

3. **CLICK** Press Esc or click outside to dismiss

→ Popover closes; drawer remains open

Tip: AI accept/reject is recorded to the audit trail (recordAiOutcome) — reviewers see who accepted what and when.

4.6 AI · Draft CAPA RCA

What it does	Larger drawer-in-drawer that produces a complete RCA: 5-Why + Fishbone + corrective + preventive + effectiveness check + regulatory clauses + citations.
Menu / route	View + AI drawer · button labelled 'Draft CAPA RCA'
Available to roles	any with drawer access
Backend API	POST /api/ai/capa/draft-rca
AI assistance	AI Free Gemini. Returns severity classification + structured RCA. User can edit every field before accepting.

CLICK-BY-CLICK

1. **CLICK** Click 'Draft CAPA RCA'

→ Full-width drawer opens with header 'AI-drafted RCA' + provider/model/latency tags

2. **WAIT** Wait ~3-5 s

→ 5-Why list renders, then corrective/preventive sections fill in, regulatory clauses appear as chips

3. **TYPE** (Optional) Edit any field — RCA narrative, action descriptions, due-day estimates

→ Edits update the local copy; the AI draft is kept for diff

4. **CLICK** Click 'Accept' (left of Create CAPA)

→ Records USER_ACCEPTED or USER_EDITED outcome in the audit trail. Does NOT create a CAPA on its own.

5. **CLICK** Click 'Create CAPA from this RCA' (primary blue, right)

→ Green Alert: 'CAPA CAPA-2026-NNNN created in CAPA_OPEN state. Open the CAPA workspace at /buyer/capas to triage and assign actions.'

walkthrough/05-kenji.png

Tip: The 'Create CAPA from this RCA' button is the cross-module hand-off. It posts to POST /api/quality/capa-from-deviation which mints a real capa-v2 record + writes the corrective/preventive actions onto a capa-v2-action-plans row + pushes the CAPA id into the deviation's linkedCAPAids.

4.7 AI · Predictive CAPA badge

What it does	Auto-renders in the View + AI drawer. Predicts P(on-time) and P(effective) for the (potential) CAPA, plus the top contributing factors.
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Menu / route	View + AI drawer · 'Predictive outcome' card at the bottom
Available to roles	any with drawer access
Backend API	POST /api/ai/predict/capa-outcome
AI assistance	AI Heuristic model (capa.heuristic@1.0.0) — features: severity, rootCauseDepth, actionCount, daysToDue.

CLICK-BY-CLICK

1. **WAIT** Auto-renders 700 ms after the drawer opens
→ Two chips: P(on-time)=81% (green if >70%) and P(effective)=66% (orange if 50-70%)
2. **WAIT** Below: Top factors line
→ e.g. 'Top factors: slack_days · owner_prior_closure_rate'

4.8 Workflow button · Assess

What it does	Captures the impact assessment (4 fields). Moves REPORTED → UNDER_ASSESSMENT.
Menu / route	/nonconformance row · 'Assess' button (visible when status = REPORTED or UNDER_ASSESSMENT)
Available to roles	EDITOR_ROLES
Backend API	POST /api/deviations/:id/assess

CLICK-BY-CLICK

1. **CLICK** Click 'Assess' on the row
→ Dialog opens with 4 fields
2. **TYPE** Fill 'Product Quality Impact'
→ free-text
3. **TYPE** Fill 'Patient Safety Impact'
→ free-text
4. **CLICK** Choose 'Batch Disposition' (RELEASE/REJECT/REWORK/REPROCESS/QUARANTINE/PENDING)
→ PENDING is the safe default if undecided
5. **TYPE** Fill 'Regulatory Impact'
→ free-text — basis for FAR / MDR decision later
6. **CLICK** Click 'Submit'
→ Dialog closes; row status chip flips to UNDER_ASSESSMENT

4.9 Workflow button · Investigate

What it does	Captures the root-cause analysis. Moves UNDER_ASSESSMENT → UNDER_INVESTIGATION; if rootCause is set → PENDING_DISPOSITION.
Menu / route	/nonconformance row · 'Investigate' button (visible when status ≤ UNDER_INVESTIGATION)
Available to roles	EDITOR_ROLES
Backend API	POST /api/deviations/:id/investigate

CLICK-BY-CLICK

1. **CLICK** Click 'Investigate'
→ Dialog opens
2. **CLICK** Choose 'Method' — FIVE_WHY / FISHBONE / FAULT_TREE / PARETO / BRAINSTORM / OTHER
→ Method captured for QA-trail
3. **TYPE** Fill 'Summary' (investigation summary)
→ free-text
4. **TYPE** Fill 'Root Cause' (one sentence)
→ Setting this advances state to PENDING_DISPOSITION
5. **CLICK** Choose 'Root Cause Category' — HUMAN_ERROR / EQUIPMENT_FAILURE / MATERIAL_DEFECT / PROCESS_GAP / etc.
→ 10 enums

6. **CLICK** Click 'Submit'
→ Status flips. If you used the AI 5-Why scaffolder first, paste its 5th-Why answer here.

4.10 Workflow button · Dispose

What it does	Records the batch disposition decision. Moves PENDING_DISPOSITION → PENDING_CAPA_DECISION.
Menu / route	/nonconformance row · 'Dispose' button (visible when status = PENDING_DISPOSITION)
Available to roles	EDITOR_ROLES (Head of QA in practice)
Backend API	POST /api/deviations/:id/dispose

CLICK-BY-CLICK

1. **CLICK** Click 'Dispose'
→ Dialog opens
2. **CLICK** Choose 'Disposition Decision' — RELEASE / REJECT / REWORK / REPROCESS / QUARANTINE / NOT_APPLICABLE
→ REJECT for the failed batch in the demo
3. **TYPE** Fill 'Justification'
→ Cite the failing test + spec ref
4. **CLICK** Click 'Submit'
→ Status flips to PENDING_CAPA_DECISION

4.11 Workflow · CAPA decision

What it does	Decide whether a systemic CAPA is required. Moves PENDING_CAPA_DECISION → CAPA_REQUIRED or PENDING_CLOSURE.
Menu / route	POST /api/deviations/:id/capa-decision (no dedicated UI button today – fired by the AI 'Create CAPA' button or via API)
Available to roles	EDITOR_ROLES
Backend API	POST /api/deviations/:id/capa-decision

CLICK-BY-CLICK

1. **API** Frontend wiring: when user clicks 'Create CAPA from this RCA' the cross-module endpoint sets capaRequired=true + populates linkedCAPAIds.
→ Status moves to CAPA_REQUIRED
2. **API** Backend body: { capaRequired: true | false, autoCreateCapa: true | false, linkedCAPAIds: [...] }
→ If autoCreateCapa=true and the deviation has no linked CAPA, the legacy code-path mints one via crossModuleService.createCapaFromDeviation()

Tip: There's no standalone 'Decide CAPA' button on the row today — the user makes the decision implicitly via the AI 'Create CAPA' button, or via API.
Roadmap: dedicated drawer.

4.12 Workflow button · Close

What it does	Final closure. Captures closureNotes + sets closedBy + closedAt. Moves PENDING_CLOSURE / CAPA_REQUIRED → CLOSED.
Menu / route	/nonconformance row · 'Close' button (visible when status = PENDING_CLOSURE or CAPA_REQUIRED)
Available to roles	EDITOR_ROLES (Head of QA / VP Quality)
Backend API	POST /api/deviations/:id/close

CLICK-BY-CLICK

1. **CLICK** Click 'Close'
→ Dialog opens
2. **TYPE** Fill 'Closure Notes' — what was done; reference to linked CAPA closure

→ Required

3. **CLICK** Click 'Submit'
- Status flips to CLOSED. closedBy + closedAt persisted. Row chip turns green.

Tip: Closure does not currently force the linked CAPA(s) to be closed first — they continue independently in the CAPA workspace. This is intentional (CAPAs may take 90+ days, but the deviation can close once disposition is final).

4.13 Cross-feature · Signal cluster alert

What it does	Z-score trend detector running across deviations. If 3+ deviations cluster on the same equipment / material / process within the baseline window, an alert fires.
Menu / route	Head-of-QA dashboard + GET /api/ai/signals?status=open
Available to roles	admin · Head of QA
Backend API	GET /api/ai/signals?status=open
AI assistance	AI Wave 3 statistical detector (no LLM).

CLICK-BY-CLICK

1. **NAVIGATE** James (Head of QA) opens the AI signals dashboard
- 1 cluster shown today: equipment:NVX-PRESS-001 · size 3 · z=3.4. Members listed by deviationNumber.

5. Full lifecycle walkthrough — one transaction, end-to-end

One deviation walked end-to-end. Each row is one transition. The 'Observed' column is what we got from the live Vercel run on 2026-04-25. Screenshot rows under each step show what the user actually saw at that point.

#	Persona	State transition	Action	API	Observed	Result
1	Kenji QA Specialist	— → REPORTED	Open /nonconformance, click '+ Report Deviation', fill the intake form (title + description + classification=MAJOR + product), Save	POST /api/deviations	id assigned · deviationNumber DEV-2026-NNNN · status=REPORTED	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Admin

Nonconformance

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and closure

+ Report Deviation

3 total

3 open

Classification

Status

Category

Viable contamination excursion · Grade C airlock (CA-2) [DEV-DEMO-003]

· UNPLANNED

MINOR

UNDER INVESTIGATION

View + AI

Investigate

Calibration drift on Korsch XL-400 tablet press during OQ [DEV-DEMO-002]

· UNPLANNED

MINOR

UNDER ASSESSMENT

View + AI

Assess

Investigate

OOS dissolution on batch NVX-2026-B014 (Novexolimus 1 mg tablet) [DEV-DEMO-001]

· UNPLANNED

MINOR

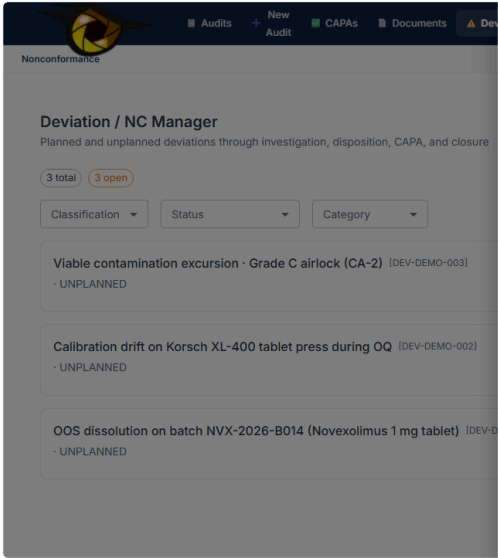
UNDER INVESTIGATION

View + AI

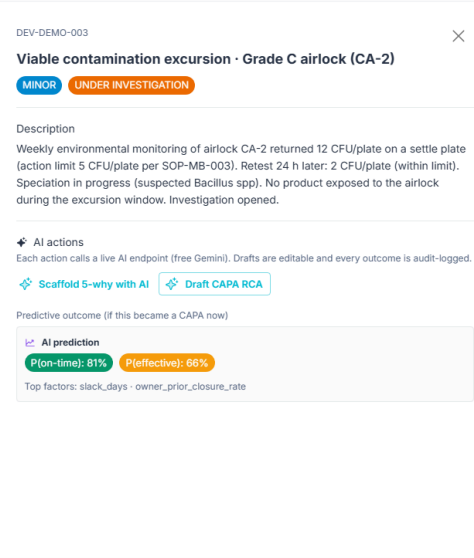
Investigate

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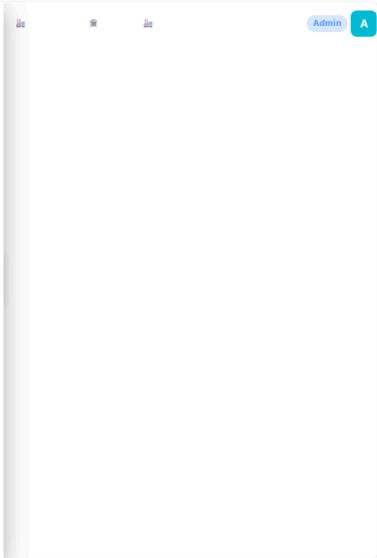
#	Persona	State transition	Action	API	Observed	Result
2	Kenji QA Specialist	REPORTED → UNDER_ASSESSMENT	Click 'Assess' on the new row → fill 4 impact fields → Submit	POST /api/deviations/:id/assess	status=UNDER_ASSESSMENT	PASS
3	Kenji QA Specialist	UNDER_ASSESSMENT → UNDER_INVESTIGATION	Click 'View + AI' on the row → Click 'Scaffold 5-why with AI' → AI returns 5-Why chain in ~3 s	POST /api/ai/deviation/scaffold-five-why	5 whys + 6 follow-ups + 6M categorisation. Provider gemini-2.5-flash-lite. Confidence 0.85.	PASS



walkthrough/03-kenji.png

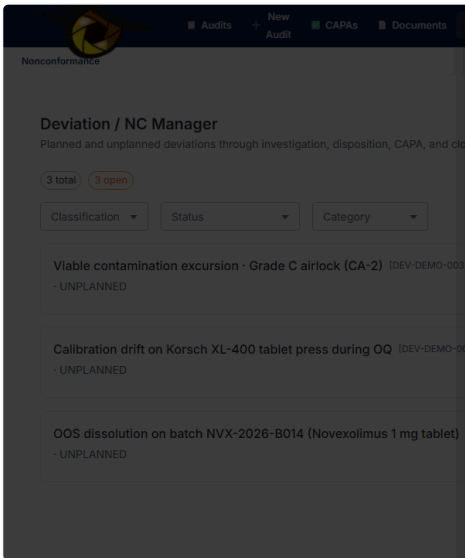


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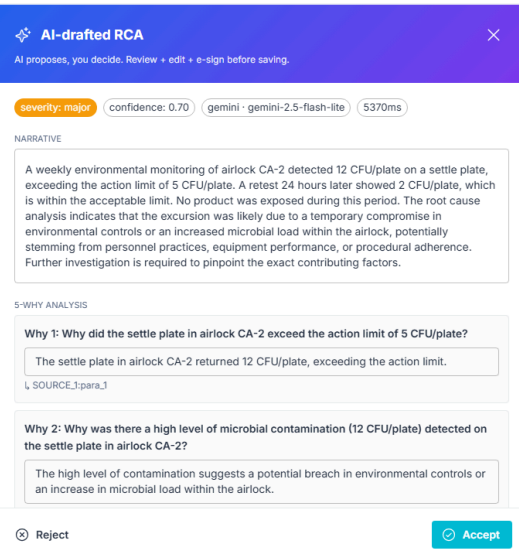


walkthrough/03-kenji.png

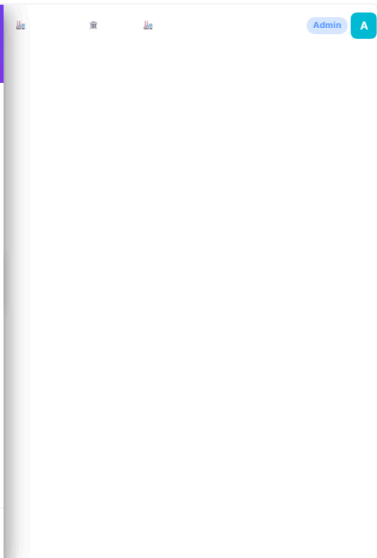
4	Kenji QA Specialist	UNDER_INVESTIGATION → PENDING_DISPOSITION	Click 'Draft CAPA RCA' in the same drawer → AI returns full RCA in ~4 s → Click 'Create CAPA from this RCA'	POST /api/ai/capa/draft-rca · then POST /api/quality/capa-from-deviation	Green alert: 'CAPA CAPA-2026-NNNN created in CAPA_OPEN state.' deviation.linkedCAPAids[] now contains the new CAPA id; capaRequired=true	PASS
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walkthrough/05-kenji.png



walkthrough/05-kenji.png



walkthrough/05-kenji.png

#	Persona	State transition	Action	API	Observed	Result
5	(verify) —	— → —	Open /buyer/capas — confirm the CAPA workspace shows the newly minted record	GET /api/capa-v2/capas	CAPA visible at top of list with status CAPA_OPEN, severity HIGH (mapped from MAJOR), owner=Kenji, dueDate=+30 days	PASS
6	James Head of QA	PENDING_DISPOSITION → PENDING_CAPA_DECISION	James opens the deviation row → Click 'Investigate' (sets rootCause from the AI draft) → Click 'Dispose' → choose REJECT + justification → Submit	POST /api/deviations/:id/investigate · POST /api/deviations/:id/dispose	status=PENDING_CAPA_DECISION	PASS
7	James Head of QA	PENDING_CAPA_DECISION → CAPA_REQUIRED	Confirm CAPA decision = required (already set when Kenji clicked 'Create CAPA from RCA')	POST /api/deviations/:id/capa-decision	status=CAPA_REQUIRED, linkedCAPAIds intact	PASS
8	James Head of QA	CAPA_REQUIRED → CLOSED	Click 'Close' on the row → fill closureNotes referencing the CAPA → Submit	POST /api/deviations/:id/close	status=CLOSED · closedBy=James · closedAt=now · linked CAPA continues independently in /buyer/capas	PASS
9	Maria Lead Auditor (audit-time)	CLOSED → CLOSED (verified)	(Audit-time) Maria opens the CAPA-v2 workspace → walks the linked CAPA through INVESTIGATION → RCA → ACTION_PLAN → IMPLEMENTATION → EFFECTIVENESS_REVIEW → CLOSED_EFFECTIVE	POST /api/capa-v2/capas/:id/* (multiple)	Covered in eqms-lifecycle.spec.ts capa-v2 test; not duplicated here	SKIP

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
5-Why Scaffolder (Wave 1)	UNDER_INVESTIGATION	POST /api/ai/deviation/scaffold-five-why	View + AI drawer · 'Scaffold 5-why with AI' button → popover	5-level Why chain with citations + 6 follow-up questions + 6M categorisation	Free Gemini 2.5 Flash-Lite
CAPA RCA Drafter (Wave 1)	UNDER_INVESTIGATION → CAPA_REQUIRED	POST /api/ai/capa/draft-rca	View + AI drawer · 'Draft CAPA RCA' button → drawer-in-drawer	Full RCA: 5-Why + Fishbone + corrective + preventive + effectiveness + regulatory clauses + citations + severity	Free Gemini 2.5 Flash-Lite
Predictive CAPA Outcome (Wave 3)	any (drawer auto-render)	POST /api/ai/predict/capa-outcome	View + AI drawer · 'Predictive outcome' card (auto-renders)	P(on-time) + P(effective) + top contributing factors	Heuristic model capa.heuristic@1.0.0 (no LLM)

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Cross-module CAPA Creator	UNDER_INVESTIGATION → CAPA_REQUIRED	POST /api/quality/capa-from-deviation	AI Draft CAPA RCA drawer · 'Create CAPA from this RCA' button (primary blue, right)	Mints a real capa-v2 record + persists corrective/preventive actions + links the new CAPA id back into the deviation	Cross-module service (no LLM)
Signal cluster detector (Wave 3)	all states (post-create)	GET /api/ai/signals?status=open	Head-of-QA dashboard	Z-score detector across deviations on shared features (equipment, material, process). Today: 1 alert on equipment:NVX-PRESS-001 · z=3.4	Statistical (no LLM)

7. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · deviation	REPORTED → CLOSED — 8 steps via API	PASS	8/8 PASS — eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F1	Deviation → AI Draft RCA → real CAPA created in workspace → linkedCAPAIds updated → close deviation	PASS	7/8 PASS, 1 SKIP (full CAPA closure deferred to v2 lifecycle)
novex-walkthrough.spec.ts	/nonconformance UI rendering with 3 seeded rows + View+AI drawer + AI buttons	PASS	screenshots 02 + 03 + 05 in walkthrough.json
Wave 1 smoke	POST /api/ai/deviation/scaffold-five-why · POST /api/ai/capa/draft-rca	PASS	11/12 wave1-3 smoke
Wave 3 smoke	POST /api/ai/predict/capa-outcome · GET /api/ai/signals	PASS	wave1-3 smoke green

8. Known gaps + roadmap

- **Dedicated 'CAPA decision' drawer (PENDING_CAPA_DECISION → CAPA_REQUIRED)** — Today the decision is implicit via the AI 'Create CAPA' button. Add an explicit Yes/No drawer with rationale field. **MEDIUM**
- **E-signature requirement on close** — Closure currently captures closedBy + closedAt but doesn't force /api/electronic-signatures/sign. Wire the signature gate for CRITICAL severity. **HIGH**
- **Field Alert Reportable flag + deadline timer** — Per 21 CFR 314.81 / 21 CFR 803, a CRITICAL deviation with patient-safety exposure has a 3-day FAR or 5-day MDR clock. Add isFieldAlertReportable boolean + farDueDate auto-computed, with reminder notification policy. **HIGH**
- **Surface linkedCAPAIds in the deviation detail drawer** — The deviation has the link in the model (verified by API) but the drawer doesn't render the linked CAPAs as clickable chips yet. **MEDIUM**
- **AI Complaint-triage agent (Wave 3 roadmap)** — Not in Deviation module per se but extends the cross-module flow to Complaint → Deviation → CAPA. **LOW**
- **ROI / token-budget tool for tenant admin** — User-requested separately. Per-tenant LLM token usage dashboard with provider override (Gemini Free / GPT-4o / Anthropic / open-source on-prem) and budget cap. **DEFERRED**